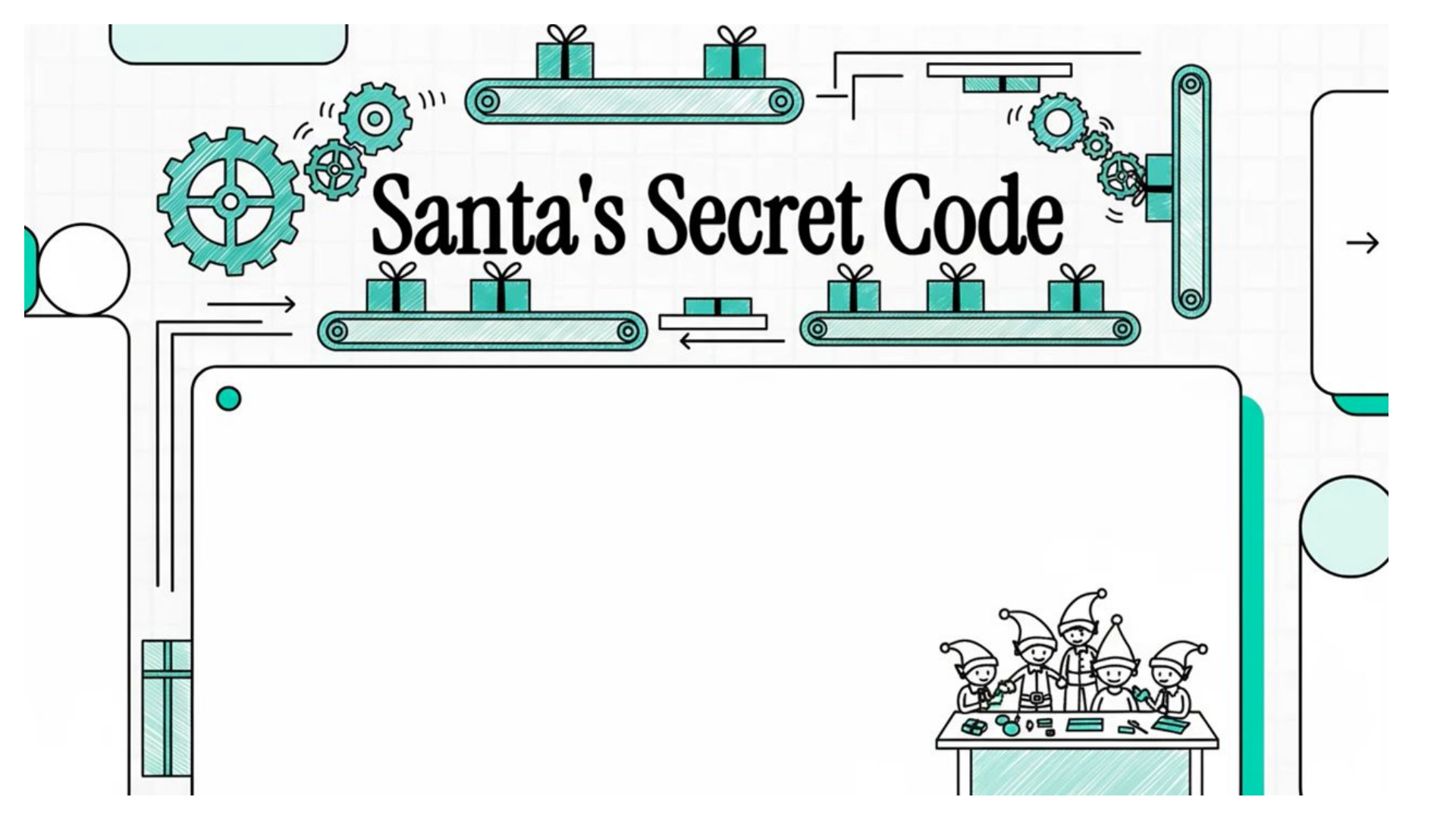


Week 12

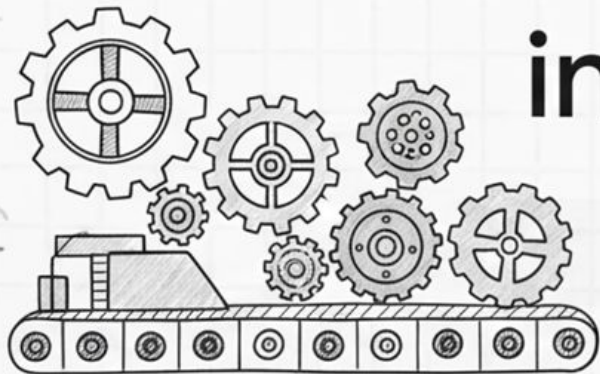
Structures & Classes

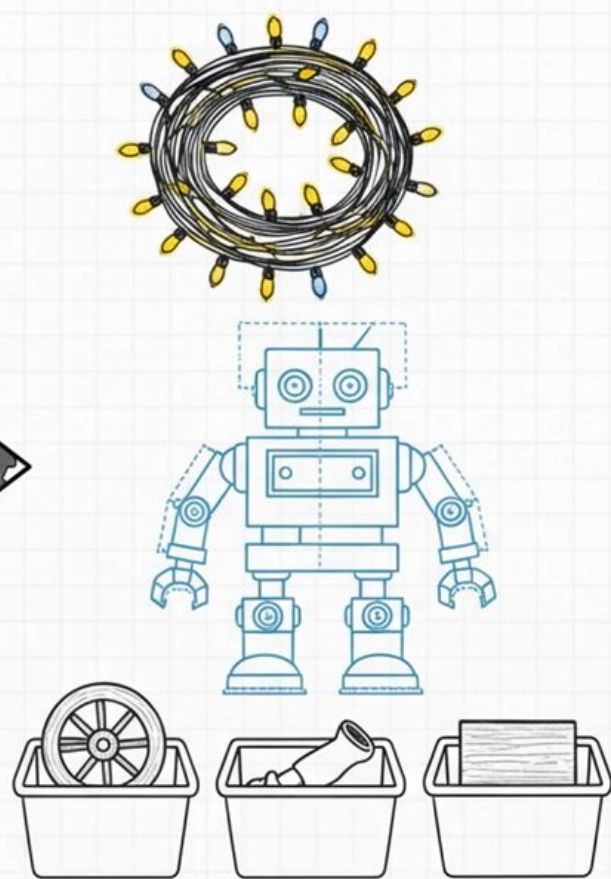
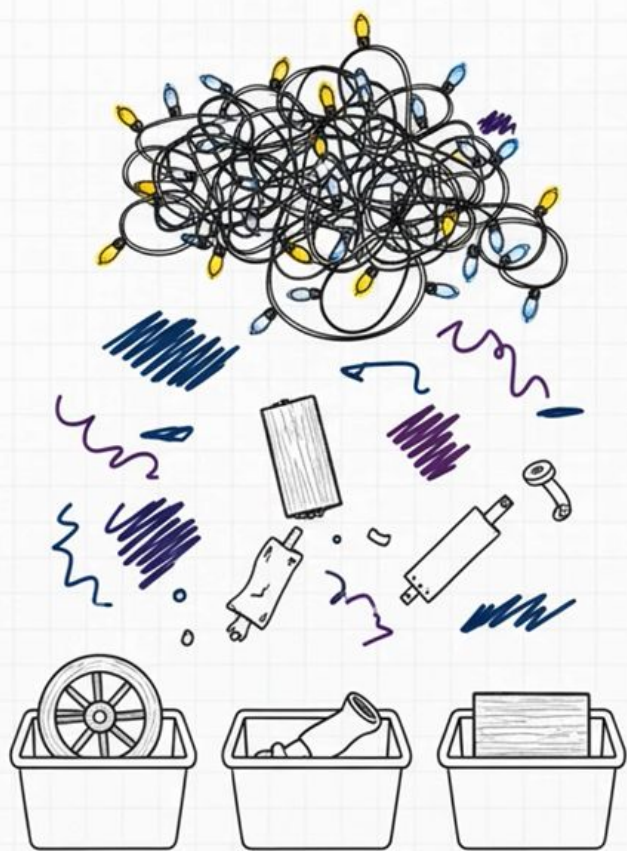


Santa's Secret Code



How does Santa build
millions of **unique toys**
in one year?



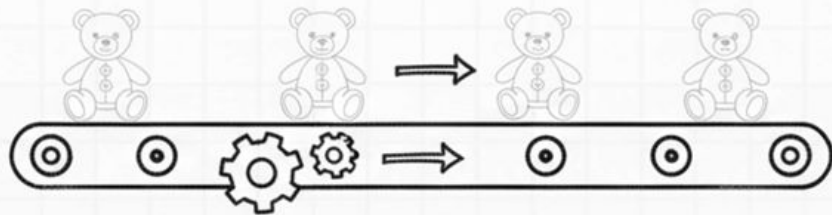




1

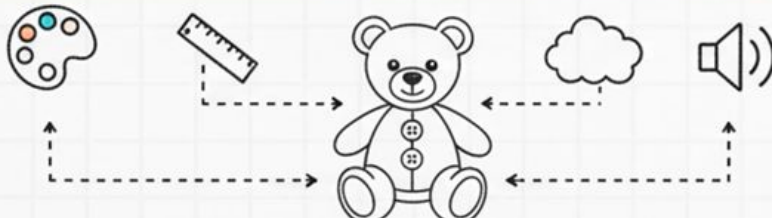
The Magic Blueprint

Classes and Objects



Class

The official blueprint for a toy. It defines what the toy has and what it can do.

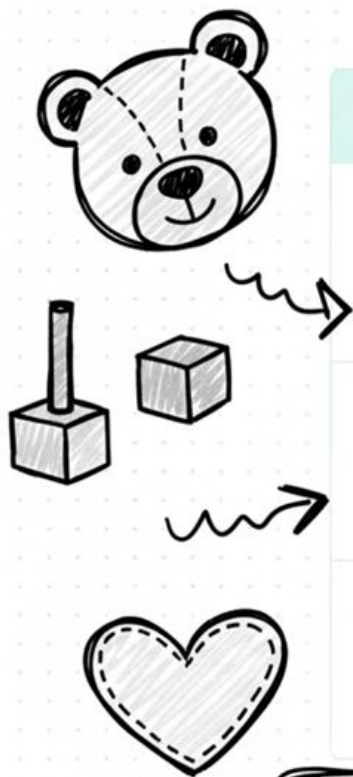


Object

The actual, physical toy built from the blueprint
Millions of toy objects can be made from one
class.







Feature (Attribute)	Action (Method)
Color: Brown	PlaySound()
Size: Medium	Hug()
Stuffing: Cotton	WaveArm()



2

The Elves' Secret Rules

The 4 Pillars of OOP





“Keep the magic **safe!**
Don't let anyone mess
with the toy's **design.**”



Encapsulation

1. Keep `private`

The core design of the toy is hidden from direct access.



2. Offer `public` tools

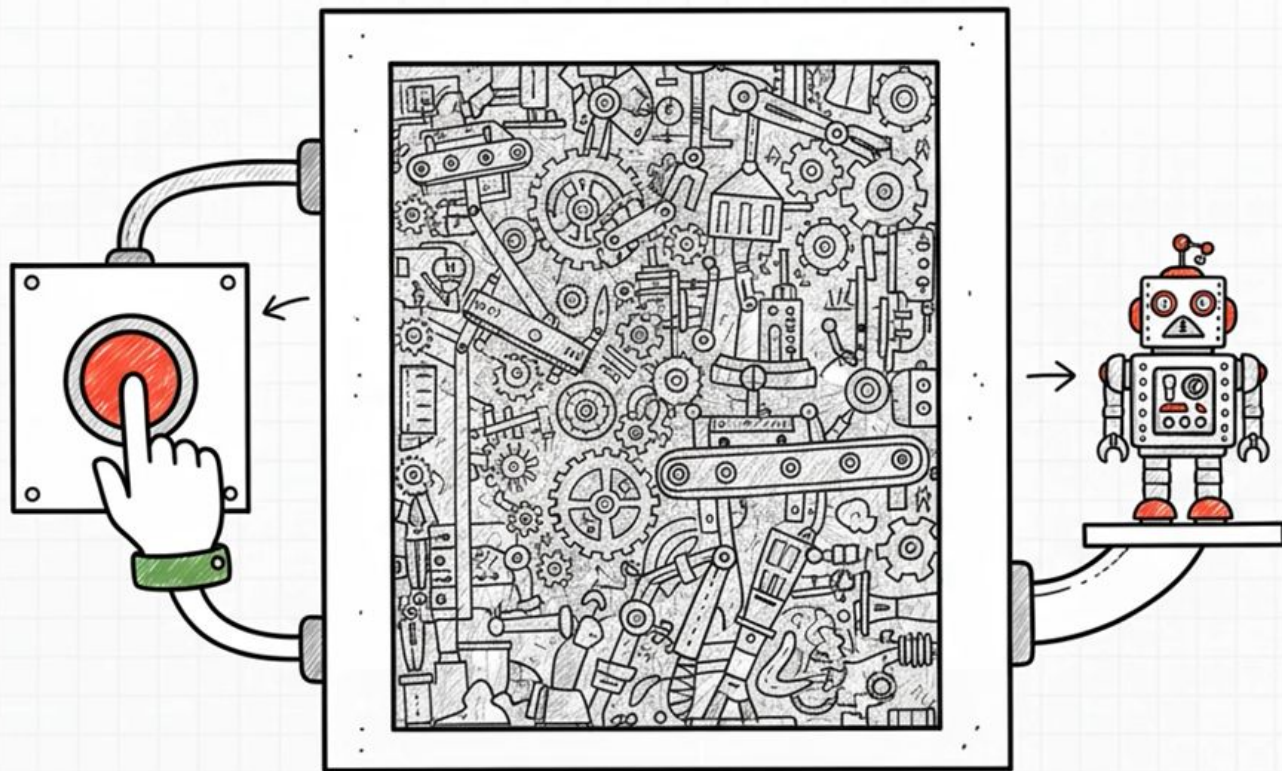
Provide safe, approved methods to interact with the design.

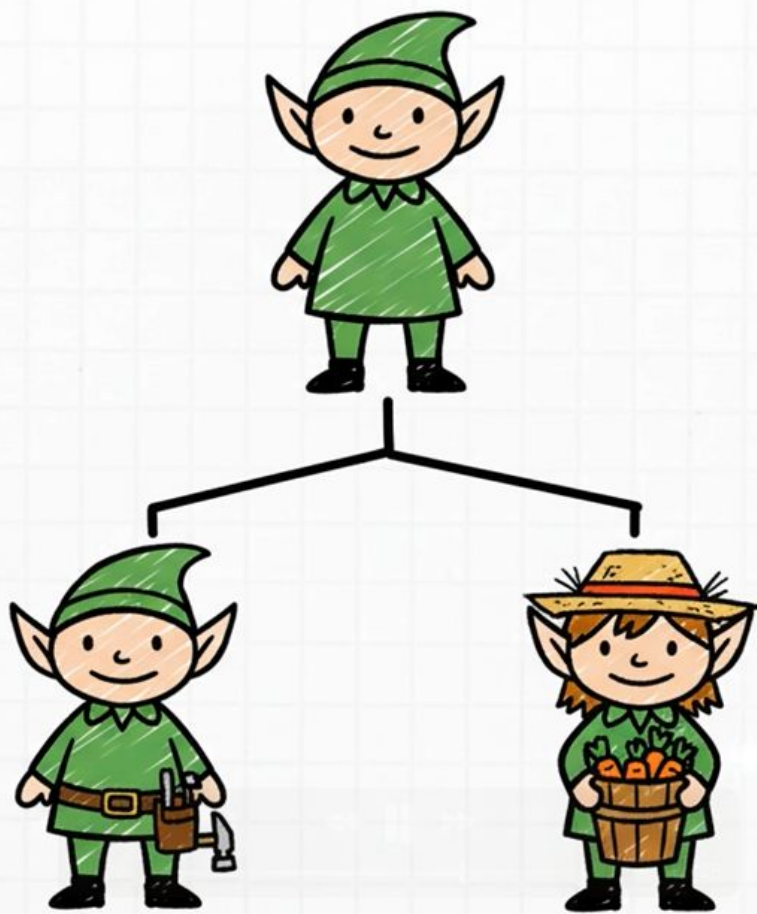


3. Protect the magic

This prevents unauthorized changes and costly mistakes.







Toy maker

Reindeer Handler



+





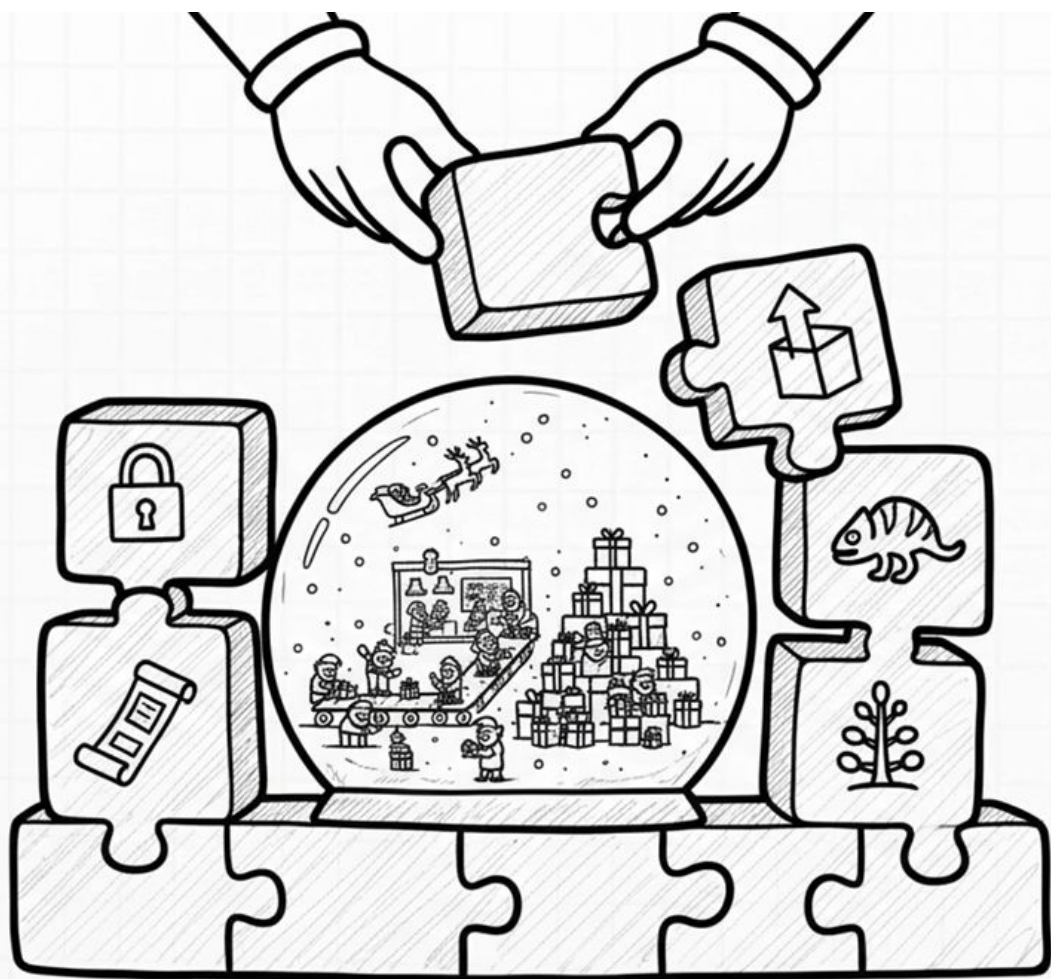
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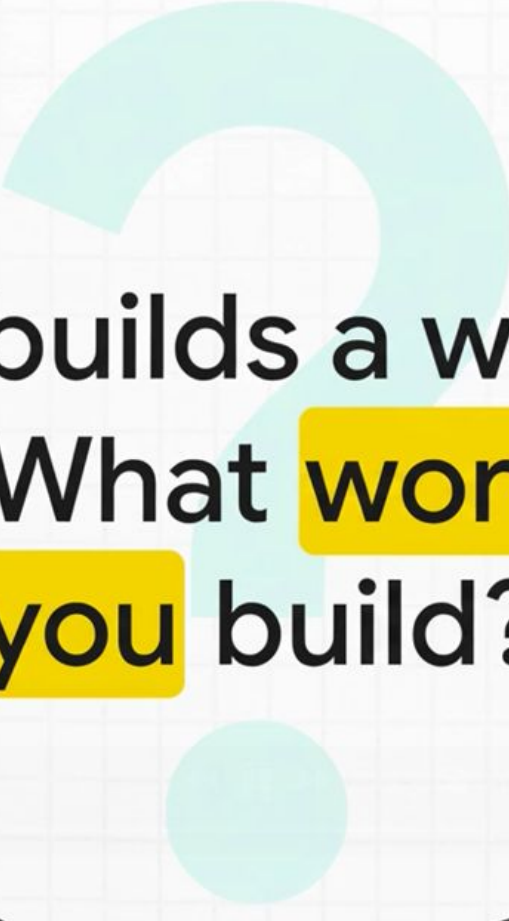
A World of Wonder

Why This Matters

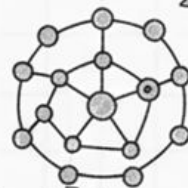
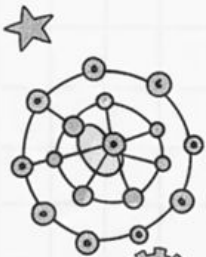
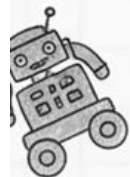








Santa builds a world of
toys. What **world**
you build?





Lab: OOP

<https://github.com/TTPR-LaGuardia-Community-College/MAC101-Lab-OOP>

Attendance